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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Time: 03 Hours

Code: 100112 ***Subject: Workshop Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) Which of the following is NOT a common sheet metal material listed?
(i) Aluminium (iii) Stainless steel
(ii) Copper (iv) Tin plate

(b) Which tool is primarily used to measure the thickness of sheet metal?
(i) Hammer (iii) Scriber Punch
(ii) Sheet metal gauge (iv) None of these

(c) Which of the following is a product mentioned under product development?
(i) Cylindrical container (iii) Hexagonal box with cap
(ii) Rectangular duct (iv) Tapered rod

(d) Which joining method typically uses a filler metal and operates below 450°C?
(i) Arc welding (iii) Soldering
(ii) Brazing (iv) Thread cutting

(e) Which type of sand is typically used for general molding purposes and has moisture content?
(i) Loam sand (iii) Dry sand
(ii) Green sand (iv) Facing sand

(f) What is the function of a sprue pin in foundry work?
(i) To shape the mold (iii) To vent
(ii) To create a passage for molten metal (iv) To hold the pattern in place

(g) Which of the following is a method of filing used in fitting shops?
(i) Upward filing (iii) Cross filing
(ii) Circular filing (iv) Reverse filing

(h) What is the main purpose of using a scriber in the fitting process?
(i) To cut metal (iii) To mark lines on metal surfaces
(ii) To hold the job (iv) To measure thickness

(i) Which of the following is a hot working process used in the smithy shop?
(i) Filing (iii) Drop forging
(ii) Cold drawing (iv) Thread cutting

(j) Which operation involves lengthening a workpiece by reducing its cross-section?
(i) Upsetting (iii) Drawing
(ii) Bending (iv) Swaging

Q.2 Solve both questions : **[14]**

(a) Explain the process of scribing in sheet metal work and its importance.

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(b) List any five tools used in sheet metal working and describe their functions briefly.

Q.3 Solve both questions : **[14]**

(a) Describe the steps involved in manufacturing a funnel using sheet metal.

(b) Explain the process of arc welding and list the safety precautions to be followed during the operation.

Q.4 Solve both questions : **[14]**

(a) List out different types of patterns used in foundry practice? Explain any three with advantages.

(b) Explain the steps involved in sand casting, starting from pattern preparation to cleaning of the final casting.

Q.5 Solve both questions : **[14]**

(a) Describe various types of molding sands and mention their properties and uses.

(b) List and describe the functions of any six tools used in pattern making and foundry practice.

Q.6 Solve both questions : **[14]**

(a) Describe the various types of files and their uses in fitting practice.

(b) Explain the safety precautions that must be observed in a fitting shop. Also explain the difference between cross filing and draw filing.

Q.7 Solve both questions : **[14]**

(a) Describe the various forging operations performed in a smithy shop.

(b) Explain the differences between hot working and cold working processes with examples.

Q.8 Solve both questions : **[14]**

(a) Discuss the process of seasoning of wood. What are the different methods of seasoning and why is it important in carpentry work?

(b) Describe the various tools used in carpentry and their specific functions.

Q.9 Write short notes on any two of the following: **[14]**

(a) Main tools and equipment used in smithy
(b) Wood working Machines
(c) Gas Welding
(d) Brazing

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